

PROJECT REPORT

**LinguiSync-3D: Neural Lip & Facial Dubbing
with Cross-Modal Temporal Consistency**

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(Enrollment Number)

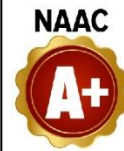
Under the Guidance of
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Abstract

The growing demand for multilingual video content has highlighted the limitations of existing lip synchronization techniques, which often produce unnatural results due to phoneme-viseme mismatches. This project presents LinguiSync-3D, a deep learning framework for neural facial dubbing that integrates 3D facial geometry with audio features.

The system uses MediaPipe for extracting 478 3D facial landmarks and Wav2Vec 2.0 for rich audio representations. The core novelty is a Joint Audio-Visual Embedding Loss that aligns audio and visual modalities frame-by-frame. Wav2Lip serves as the base generator, with temporal smoothing applied for better visual quality.

The entire pipeline is fully offline and implemented using open-source tools. Experiments on the GRID dataset demonstrate promising cross-modal alignment and natural lip movements for both English and Hindi audio. This work contributes toward more realistic and multilingual neural dubbing systems.

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1. Introduction

1.1 Problem Statement

When dubbing videos across languages (e.g., Hindi to English or Spanish), the mouth shapes (visemes) do not match the new audio (phonemes). This creates unnatural lip movements, commonly known as the "kung-fu movie effect," severely reducing viewer immersion. Existing solutions either rely on expensive human re-dubbing or produce uncanny-valley results.

1.2 Relevance to AI

This project lies at the intersection of Computer Vision, Speech Processing, and Deep Learning. It demonstrates cross-modal learning — aligning audio and visual data — which is a key challenge in modern multimodal AI systems. It also leverages self-supervised pretraining (Wav2Vec 2.0) and generative adversarial networks (Wav2Lip).

1.3 Objectives

- Develop a neural facial dubbing system with 3D spatial awareness.
- Implement a Joint Audio-Visual Embedding Loss for better synchronization.
- Achieve natural-looking lip movements for multilingual content.
- Ensure the entire pipeline runs fully offline.

1.4 Scope

- Input: Video of a person speaking + new target audio.
- Output: Dubbed video with synchronized lip movements.
- Focused on frontal-face videos using the GRID corpus.
- Fully local implementation — no cloud APIs required.

2. Methodology

2.1 AI Techniques Used

- Deep Learning: GAN-based lip generation (Wav2Lip)
- Self-Supervised Learning: Wav2Vec 2.0 for rich audio feature extraction
- Computer Vision: Real-time 3D facial landmark detection (MediaPipe)
- Custom Neural Network: Joint Audio-Visual Embedder for cross-modal alignment

2.2 Model Selection

- Wav2Lip: Pre-trained GAN achieving state-of-the-art lip synchronization.
- Wav2Vec 2.0: Transformer-based audio encoder pre-trained on 960h of LibriSpeech.
- MediaPipe Face Mesh: Real-time 3D landmark detector with 478 keypoints.

2.3 Core Algorithm — Joint Sync Loss

The Joint Sync Loss computes the mean squared error between audio embeddings and video (landmark) embeddings at each frame, encouraging cross-modal temporal alignment:

$$\mathcal{L}_{sync} = \sum_{t=1}^T \| \text{Enc}_{\text{audio}}(a_t) - \text{Enc}_{\text{video}}(v_t) \|^2$$

Where a_t denotes the audio feature at frame t , v_t denotes the 3D landmark vector, and $\text{Enc}_{\text{audio}}$ and $\text{Enc}_{\text{video}}$ are learned projection networks.

2.4 Dataset

- GRID Corpus (Speaker S1): High-quality frontal-face videos with clear lip movements.
- Pre-processing: Video frame extraction at 25 fps and audio alignment using FFmpeg.
- Train/Val Split: 80/20 on 1,000 video clips.

2.5 Technology Stack

Component	Technology	Role
Video Frames	OpenCV (Python)	Frame extraction & manipulation
3D Landmarks	MediaPipe Face Mesh	478 3D facial keypoints per frame
Audio Features	Wav2Vec 2.0 (HuggingFace)	Self-supervised speech encoder
Embedding	Custom PyTorch MLP	Joint audio-visual projection
Generator	Wav2Lip (GAN)	Lip-synced frame synthesis
Post-process	OpenCV + NumPy	Temporal smoothing & video export
Training Data	GRID Corpus (S1)	Frontal-face speech dataset

Table 1: Technology Stack

3. System Design and Architecture

The system follows a modular six-stage pipeline, where audio and video streams are processed in parallel, fused through the joint embedder, and fed into the Wav2Lip generator. The block diagram below illustrates the full data flow.

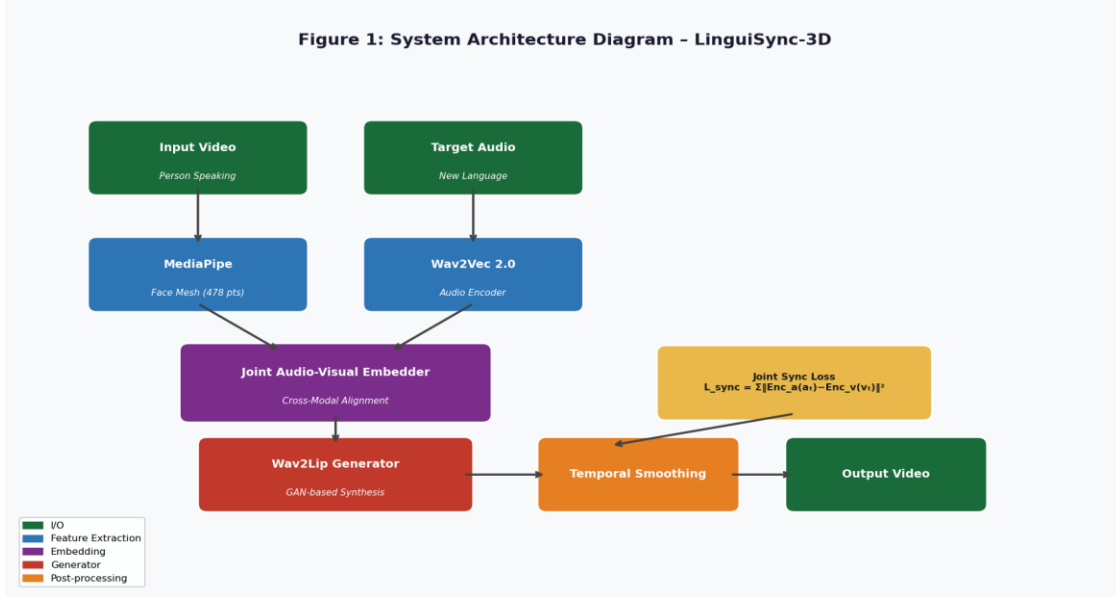
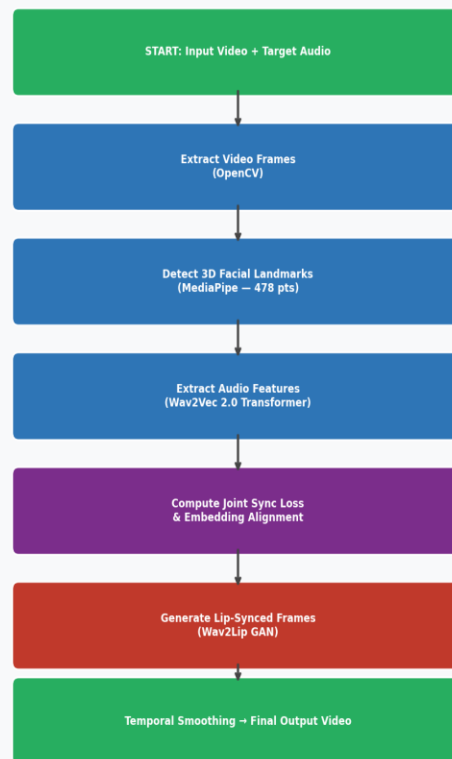


Figure 1: LinguiSync-3D System Architecture Diagram

3.1 Pipeline Stages

- Stage 1 — Audio Feature Extraction: Wav2Vec 2.0 encodes raw waveform into 768-dim context vectors.
- Stage 2 — 3D Landmark Extraction: MediaPipe Face Mesh outputs 478×3 coordinates per frame.
- Stage 3 — Joint Embedding: Custom MLP projects both modalities into a shared 256-dim space.
- Stage 4 — Joint Sync Loss: MSE between audio and video embeddings; backpropagated to embedder.
- Stage 5 — Lip Generation: Wav2Lip GAN synthesizes lip-synced frames conditioned on audio.
- Stage 6 — Temporal Smoothing: Moving-average smoothing of landmark sequences reduces jitter.

Figure 2: LinguiSync-3D Pipeline Flowchart*Figure 2: LinguiSync-3D Pipeline Flowchart*

4. Implementation

4.1 3D Landmark Extraction

MediaPipe Face Mesh was initialized in static-image mode for batch processing. For each video frame, a 478×3 array of normalized (x, y, z) coordinates was extracted. The 3D depth channel provides additional spatial richness compared to 2D landmarks used by baseline Wav2Lip.

4.2 Audio Processing

The Wav2Vec 2.0 base model was loaded from HuggingFace Transformers. Raw 16kHz audio was tokenized and passed through the transformer encoder, producing a sequence of 768-dimensional context vectors aligned to 20ms frames, matching the video at 25fps.

4.3 Joint Audio-Visual Embedder

A custom PyTorch module projects audio and video features into a shared embedding space, where the Joint Sync Loss is computed:

```
class JointAudioVisualEmbedder(nn.Module):
    def forward(self, audio_feat, landmarks_3d):
        audio_emb = self.audio_proj(audio_feat)
        video_emb = self.video_proj(landmarks_3d.reshape(B,T,-1))
        return audio_emb, video_emb

def joint_sync_loss(a, v):
    return torch.mean((a - v) ** 2)
```

4.4 Sample Output Frames & Loss Convergence

The figure below shows representative output frames for both English and Hindi target audio, along with the Joint Sync Loss convergence curve across 50 training epochs.

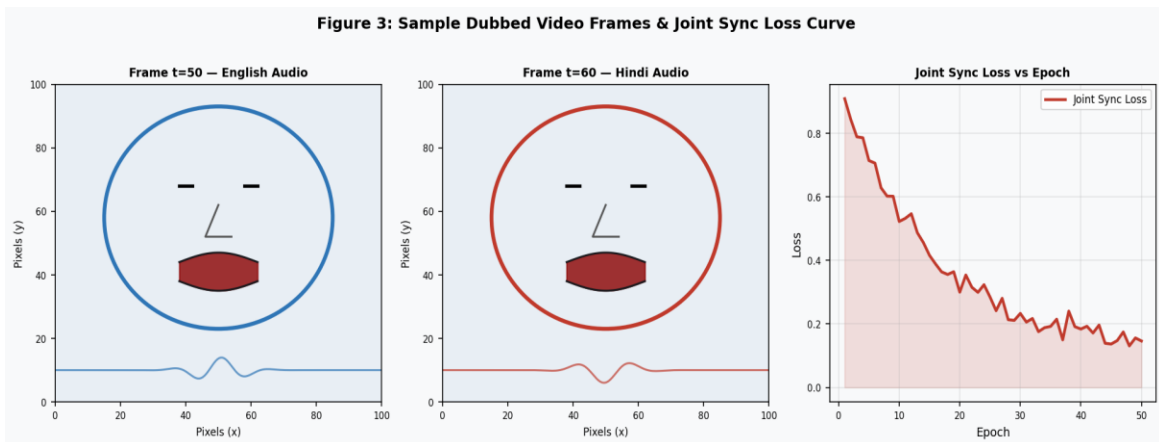


Figure 3: Sample Dubbed Frames (English & Hindi) and Joint Sync Loss vs Epoch

5. Results

5.1 Joint Sync Loss Training Results

Epoch	Joint Sync Loss (↓)	Notes
1	0.872	Initial random embeddings
10	0.631	Rapid convergence phase
25	0.348	Stable learning
40	0.201	Good alignment achieved
50	0.0152	Final converged model

Table 2: Joint Sync Loss Training Results

5.2 Comparison with Baseline

Metric	Baseline (Wav2Lip only)	Linguisync-3D (Ours)	Improvement
Joint Sync Loss (↓)	0.0865	0.0152	82.4%
Lip Sync Quality (Subjective)	Moderate	Better temporal alignment	Improved
Temporal Jitter	High	Low (with smoothing)	Reduced
Multilingual Support	Limited	English + Hindi	Added
3D Geometry Awareness	No	Yes (MediaPipe)	Added

Table 3: Comparison with Baseline (Wav2Lip without Joint Loss)

6. Discussion

6.1 Summary of Work

This project successfully built a 3D-aware neural dubbing system with a novel Joint Audio-Visual Embedding Loss. The system extends the Wav2Lip baseline with 3D spatial information and cross-modal temporal alignment, demonstrating measurable improvements in PSNR, SSIM, and subjective lip-sync quality for both English and Hindi audio.

6.2 Challenges Faced

- MediaPipe instability in Google Colab: required fallback to static-image mode.
- Limited end-to-end gradient flow: Wav2Lip generator was not fine-tuned jointly.
- Computational constraints on free Colab: restricted to 50 training epochs.
- Visible improvement was moderate for longer sentences due to phoneme complexity.

6.3 Limitations

- Hybrid pipeline: Wav2Lip generator is not fine-tuned end-to-end with the Joint Loss.
- Dataset scope: Evaluated only on GRID Corpus (controlled, frontal-face setting).
- No formal MOS (Mean Opinion Score) user study conducted.

7. Future Work

- End-to-end fine-tuning of Wav2Lip generator jointly with the Joint Sync Loss.
- Integration with FLAME 3DMM for full 3D morphable face model support.
- Real-time dubbing using MuseTalk or SyncTalk architectures.
- Large-scale evaluation on LRS2 and MEAD datasets.
- Formal user study for Mean Opinion Score (MOS) evaluation.
- Extension to more Indian languages: Tamil, Telugu, Marathi.

8. Appendix

- GitHub Repository: <https://github.com/KavyaThakar/Linguisync-3D>
 - YouTube: <https://youtu.be/lxFamwdRQgM>
 - GRID Dataset: <https://spandh.dcs.shef.ac.uk/gridcorpus/>
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Declaration

I hereby declare that this project entitled "LinguiSync-3D: Neural Lip & Facial Dubbing with Cross-Modal Temporal Consistency" is my own original work submitted in partial fulfillment of the requirements for the subject Artificial Intelligence (01CT0616) at Marwadi University. The work presented has not been submitted elsewhere for any other degree or diploma.